

Raghav Mangalapalli

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EDUCATION

Eindhoven University of Technology

Eindhoven, The Netherlands

Master of Science in Computer Science and Engineering

Sep 2025 – Present

- **Relevant Coursework:** Engineering Data-Intensive Systems; Architecture of Distributed Systems; ML Engineering; ML for Industry; Foundations of AI.

Birla Institute of Technology and Science, Pilani

Goa, India

Bachelor of Engineering in Computer Science and Master of Science in Economics

Aug 2019 – Jul 2024

EXPERIENCE

Eltropy

Jan 2024 – Jul 2025

Software Engineer - AI Team

Bangalore, India

- Engineered front-end for a conversation-intelligence insights dashboard using React and Typescript to track 2000 weekly member-agent calls, providing real time alerts and action items to agents.
- Built scalable REST APIs using Golang and MongoDB to interact with and manage configurations for 80+ LLM based conversational agents across 12+ domains.
- Designed and implemented an SFTP File Management System using Temporal and AWS S3 to sync knowledge source files for a RAG Pipeline used by Knowledge Assistants for 10+ Credit Unions.

Amazon

Jun 2023 – Dec 2023

Software Development Engineer Intern - Retail Business Services

Bangalore, India

- Built an end-to-end API integration in Java for an internal task management system using AWS Lambda, SQS, and SNS, automating workflows for Business Associates.
- Architected a serverless document conversion service using AWS Lambda Layers, Java, and TypeScript to handle multi-format file processing.
- Deployed a metrics tracking dashboard with AWS Quick Sight integrating logs from various AWS services to provide insights into the quality of the generated reports.

PROJECTS

Shapley Value based Data Valuation | *Python, NumPy, Scikit-learn*

June 2024 – Dec 2024

- Employed Shapley values as a data valuation scheme to quantify the marginal contribution of each training sample, improving dataset quality by identifying and removing noise and mislabeled points.
- Adopted KNN-Shapley to scale data valuation efficiently, reducing computational complexity from $\mathcal{O}(2^n)$ to $\mathcal{O}(n \log n)$ compared to exact Shapley computation.
- Benchmarked against naive data cleaning baselines on a medical dataset used to predict risk of diabetes, reducing dataset size by 14% while improving model accuracy by 8%.

Medical Image Classification and Interpretability | *Python, PyTorch*

Jan 2026 – Mar 2026

- Fine-tuned ResNet-18 and ViT-Base on DermaMNIST skin lesion dataset, comparing CNN vs transformer architectures across classification accuracy, training dynamics, and computational efficiency.
- Implemented GradCAM and Attention Rollout to visualize model decision-making, analyzing artifact sensitivity and lesion focus across correct, incorrect, and low-confidence predictions.
- Applied LoRA to fine-tune Gemma-3 on MedMCQA, reducing trainable parameters by ~99% while enabling accurate medical question answering compared to the base model.

TECHNICAL SKILLS

Languages & Frameworks: Golang, Python, TypeScript, C++, SQL, React, Node.js, FastAPI

Databases & Cloud: MongoDB, Redis, AWS, Docker, Temporal, Git, Grafana, Prometheus

Statistics: Time Series (ARIMA, GARCH, Unit Root Testing), Panel Data (Fixed/Random Effects, Hausman Test)

AWARDS

Winner, EMPOWER (IIT-Madras) Student Design Challenge 2022: Came first out of 90+ teams, won for designing an Arduino based chemistry lab equipment

Prof. Suresh Ramaswamy Memorial Award: Awarded annually to a student team demonstrating exceptional innovation and creativity in the fields of science/engineering. Rs. 40000 Prize Fund received.